

Mustafa Sultan

AI Engineer | Machine Learning & Deep Learning | LLM Applications

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Professional Summary

AI engineer and recent Artificial Intelligence graduate with hands-on experience building multi-agent LLM applications, RAG systems, backend APIs, and computer vision models. Experienced with LangGraph, LangChain, LangSmith, Django, FastAPI, PyTorch, and Hugging Face Transformers, with additional experience contributing to peer-reviewed machine learning research.

Technical Skills

Programming	Python, SQL, C++, C#
AI/ML	PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, Machine Learning, Deep Learning, NLP, Computer Vision
Generative AI	LLM Integration, LangGraph, LangChain, LangSmith, Agentic AI, Multi-Agent Systems, RAG, Hybrid Search, Reranking, Embeddings, Prompt Engineering, Tool Calling
Backend & Data	Django, Django Ninja, FastAPI, REST APIs, PostgreSQL, Pandas, NumPy, Pydantic
Agentic Coding	OpenAI Codex, Antigravity
Software Engineering	Git, GitHub, OOP, Data Structures and Algorithms, Version Control
Additional	Qiskit, Quantum Optimization, Competitive Programming

Education

B.Sc. in Artificial Intelligence

Nile University

Oct 2022 – Jun 2026

Egypt

- GPA: 3.82/4.00; Dean's List: Fall 2024.
- Completed and passed all academic requirements; official graduation documentation pending.

Certifications

- Machine Learning in Production** – DeepLearning.AI, Coursera (2025). Verified credential
- Generative AI with Large Language Models** – DeepLearning.AI & AWS, Coursera (2025). Verified credential

Experience

AI Engineer Intern

CrossWorkers Egypt

Jul 2025 – Sep 2025

Cairo, Egypt

- Built an AI lead-generation pipeline that analyzed email newsletters to extract job postings and company information.
- Developed a machine learning model to predict employee satisfaction from historical HR data.
- Created an HR chatbot for natural-language querying over SQL databases.
- Integrated LLM APIs and LangGraph to orchestrate agentic workflows across backend services and live databases.
- Implemented application components using Python and C# for internal AI workflows.

Research Intern – Machine Learning Classification

Nile University

Jun 2024 – Sep 2024

Giza, Egypt

- Performed preprocessing, EDA, and feature engineering on mobile traffic and POI datasets.
- Implemented and evaluated classification models for socioeconomic indicator prediction.
- Contributed to a peer-reviewed publication based on the project results.

Research Intern – Quantum Optimization

Nile University

Jun 2024 – Sep 2024

Giza, Egypt

- Applied QAOA to unit commitment optimization problems and explored quantum optimization concepts in a research setting.

Selected Projects

FGaiB – Multi-Agent Personal Finance Android Application

2025 – 2026

- Built the Python/Django backend and AI architecture for an end-to-end personal finance application; used Codex and Antigravity for agentic coding assistance on the React Native/Expo frontend.
- Engineered LangGraph multi-agent workflows that route requests to specialized financial agents while preserving user context with PostgreSQL-backed checkpoints.
- Implemented confirmation-gated tool calling, conversation memory, budget-risk alerts, goal tracking, spending analysis, and encryption for sensitive data.
- Instrumented LLM workflows with LangSmith tracing to inspect agent runs, debug failures, and improve reliability.
- Built a custom web admin dashboard for usage statistics, token monitoring, LLM cost calculation, forecasting, and anomaly detection.
- Ranked first among eight faculty graduation-project committees.

Technologies: Python, Django, Django Ninja, LangGraph, LangChain, LangSmith, PostgreSQL, Pydantic, REST APIs, React Native/Expo exposure, Codex, Antigravity

SlideTutor – AI-Powered PowerPoint Study Workspace

- Built the FastAPI backend and AI pipeline for a PowerPoint study workspace that extracts structured PPTX text, tables, images, element types, and bounding boxes while rendering slides for visual study; developed the web interface with agentic coding assistance.
- Engineered LangGraph summarizer and ReAct tutor agents with deck briefings, thread memory, and slide-aware tools for search, context, images, and assets.
- Added configurable OpenAI-compatible model support for local and cloud providers, including LM Studio, OpenRouter, OpenAI, Ollama, and vLLM.

github.com/mustafa200312/slidetutor

Technologies: Python, FastAPI, LangGraph, LangChain, OpenAI-compatible APIs, python-pptx

Al-Maktaba Al-Digital – Arabic Book-Constrained RAG System

- Built a citation-enforced RAG system over selected Arabic classical books and texts.
- Implemented Arabic document processing, chunking, embeddings, hybrid search, reranking, and grounded answer generation.
- Combined semantic and keyword retrieval, then reranked passages to improve source relevance.

Multi-View Ultrasound Imaging with a Vision Foundation Model

- Developed a two-view PyTorch model using a frozen pretrained DINOv3 backbone for longitudinal and transverse ultrasound scans.
- Built three-class segmentation for background, vessel, and plaque regions, plus binary plaque classification using both views.
- Trained with cross-entropy and Dice-based losses, handled class imbalance, and evaluated per-class Dice and F1.

Technologies: Python, PyTorch, Hugging Face Transformers, DINOv3, AdamW, Jupyter Notebook

Publication & Achievements

- *A Multi-label Classification Framework for Socioeconomic Insights Using Mobile Traffic and POI Data.* doi.org/10.1007/978-3-031-87719-3_16
- Qualified for ECPC 2023 and 2024.
- Won 2nd Place in the Linear Algebra Course category at the 17th Undergraduate Research Forum, Nile University.

Leadership, Volunteer Work & Community Engagement

Junior Teaching Assistant

Oct 2023 – Jan 2024

Nile University

- Supported students in programming, data structures, algorithms, data analysis, preprocessing, and feature engineering; explained concepts and helped debug code.

Junior Mentor / ICPC Community Mentor

Oct 2023 – Jun 2024

Nile University ICPC Community

- Mentored students in competitive programming and helped prepare weekly Codeforces problem sheets, a technical session, and ECPC registration documents.

Vice President

Sep 2024 – May 2025

University Gaming Club

- Co-organized gaming sessions and small tournaments and supported event planning and logistics.